# Juan Estanislao Hoszowski

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## Skills

**Backend**: C++, Python, TypeScript, Java (Spring Boot), Go, FastAPI, Express.js, PostgreSQL, Neo4j, MongoDB
**Frontend**: React, React Native, SDL2, CSS
**DevOps**: Docker, RabbitMQ, GitHub Actions, CI/CD, Git
**Cloud/Infra**: Render, Firebase
**Practices**: Microservices, Distributed Systems, REST APIs, System Design, Unit Testing, TDD, BDD, Agile/Scrum, OOP
**Languages**: Spanish, English, French

## Technical Projects

**Microservices Social Platform (Team of 5) | GitHub** Sep 2024 – Dec 2024
***Stack:*** *Python, TypeScript, Java, FastAPI, React Native, PostgreSQL, Neo4j, MongoDB, Docker, Firebase*

- Contributed to the development and deployment of a 12-microservice social platform covering authentication, user management, posts, real-time chat, push notifications, and backoffice services, with each service independently containerized and deployed on Render.
- Architected a polyglot persistence layer, selecting PostgreSQL for relational data, Neo4j for social-graph traversal, and MongoDB for real-time chat storage.
- Primary owner of the Twits microservice, implementing post creation, likes, reposts, comments, and personalized feed generation; also contributed to the Authentication microservice.

**Worms Remake (Team of 4) | GitHub** Oct 2023 – Dec 2023
***Stack:*** *C++, Box2D, SDL2, Qt, CMake, YAML*

- Engineered a multiplayer game engine in C++ from scratch, supporting concurrent matches with authoritative server-side physics (Box2D) to prevent client-side cheating.
- Designed a binary networking protocol with snapshot-based state sync, eliminating client-state divergence under packet loss and latency spikes.
- Shipped a full feature set — 10 weapons with unique physics behaviors, trap-based supply drops, and a Qt-based map editor.
- Primary owner of the networking layer, implementing the communication protocol and multithreaded server architecture supporting multiple simultaneous players and concurrent matches.

**Fault-Tolerant Data Processing System (Team of 3) | GitHub** Apr 2025 – Jun 2025
***Stack:*** *Python, RabbitMQ, Docker, Pandas, TextBlob*

- Built a distributed data pipeline in Python, coordinating workers via RabbitMQ to parallelize filtering, joins, and aggregation over 1GB+ CSV datasets.
- Achieved near-linear horizontal scaling: adding workers (1→3 per node) cut processing time 50% (68s→34s) under 3 concurrent clients with zero code changes.
- Removed single points of failure with a ring-based health-check protocol that auto-detects and recovers from node crashes without manual intervention.

## Education

**Universidad de Buenos Aires (UBA), Facultad de Ingeniería** Buenos Aires, Argentina
*Bachelor's Degree in Software Engineering* *Graduated Dec 2025*